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RHEOLOGY OF MONODISPERSE LATICES

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LIST OF SYMBOLS

a particle radius
a_j viscosity parameter in Ree–Eyring equation
c electrolyte concentration
c_{min} electrolyte concentration at minimum viscosity
D center-to-center particle separation; dielectric constant
D_i diameter of the ith particle
D_n number average diameter
D_s volume-to-surface average diameter
D_w weight average diameter
f function
h relative viscosity function
i index
j index
K crowding factor ($= p^{-1}$)
k Boltzmann's constant
m exponent
n particle concentration
$O(x)$ terms of the order of x

p packing fraction
q particle charge
R_i internal Reynolds number
S relative standard deviation
T absolute temperature
t time
t^* characteristic time
t_r reduced time
α undetermined number of order unity
$\dot{\gamma}$ rate of shear
$\dot{\gamma}_{(1)}$ unit rate of shear
Δ thickness of surfactant monolayer
ϵ electronic charge
η viscosity
$[\eta]$ intrinsic viscosity
η_1 limiting viscosity at high shear rate
$\eta_{(1)}$ viscosity at unit shear rate
η_2 limiting viscosity at low shear rate
η_0 viscosity of medium
η_r relative viscosity
η_{1r} limiting relative viscosity at high shear rate
η_{2r} limiting relative viscosity at low shear rate
κ ionic strength
λ_j relaxation time for jth flow unit
ρ_0 density of medium
ρ_p density of particle
ρ_r relative density
σ_D standard deviation of particle diameter
τ shear stress
τ_c characteristic shear stress in Williamson equation for relative viscosity
τ_m characteristic shear stress in Williamson equation
τ_r reduced shear stress
ϕ volume fraction
ϕ_1 volume fraction of portion 1
ϕ_{12} effective volume fraction of portion 1 in presence of portion 2
ω angular velocity

INTRODUCTION

Whereas ordinary liquids and solutions exhibit simple Newtonian flow behavior, colloidal dispersions and macromolecular solutions frequently exhibit

such rheological anomalies as non-Newtonian viscosity and time-dependent thixotropy. The common structural feature unifying this rheologically interesting class of fluids is the presence of flow units, either colloidal particles or macromolecules, whose dimensions are large compared with molecules of the suspending medium, while still small enough to exhibit significant rotatory and translatory Brownian movement. (In the case of macromolecules, internal Brownian movement may also play a significant role.) The structure of the fluid thus resembles a gas of large molecules, with the void space replaced by a Newtonian continuum. The proper framework for theoretical treatment of equilibrium and transport properties in such a system would be statistical hydrodynamics, an as yet underdeveloped field.

In the statistical-mechanical treatment of gases, a useful abstraction is the hard-sphere model. Treated both with and without an added attractive potential, this model has yielded much valuable insight into equilibrium and transport properties of real gases. The hard sphere has also found an important place in the theory of colloids, having been adopted by Einstein[1] in his study of the viscosity of dilute suspensions, by Verwey and Overbeek[2] in their classical investigation of colloid stability, and by Kirkwood and Mazur[3] to represent clustering in globular proteins. The usefulness of this approach promises to be greatly enhanced by recent developments in the production and characterization of monodisperse latices, which permit the experimentalist to approach the theoretician's ideal of a hard-sphere colloid with remarkable fidelity.

The word "latex", originally coined to denote the milky sap from certain plants such as the India-rubber tree (*Hevea brasiliensis*), has been extended to aqueous dispersions of synthetic polymers as produced by emulsion polymerization. The polymer particles in synthetic latices are near-perfect spheres, which deform negligibly under the stresses encountered in rheological studies. While the distribution of particle diameters in a typical latex is non-uniform (heterodisperse), highly monodisperse latices (Fig. 1) can be produced by special techniques of emulsion polymerization[4]. These new products, together with techniques for their purification, control of particle charge, and redispersal in non-aqueous media, make possible a renewed confrontation between theory and experiment.

The emphasis in this article is placed on monodisperse latices as model systems in the elucidation of structure–rheology relationships. The historical section which follows cites papers which contribute to this viewpoint, but is not intended as a comprehensive review of the literature. The succeeding section on the colloid chemistry of latices discusses problems which must be considered when identifying latex systems with simple model dispersions. In the next section, the structural–rheological "equation of state" is subjected to dimensional analysis, to develop a corresponding-states principle. Later

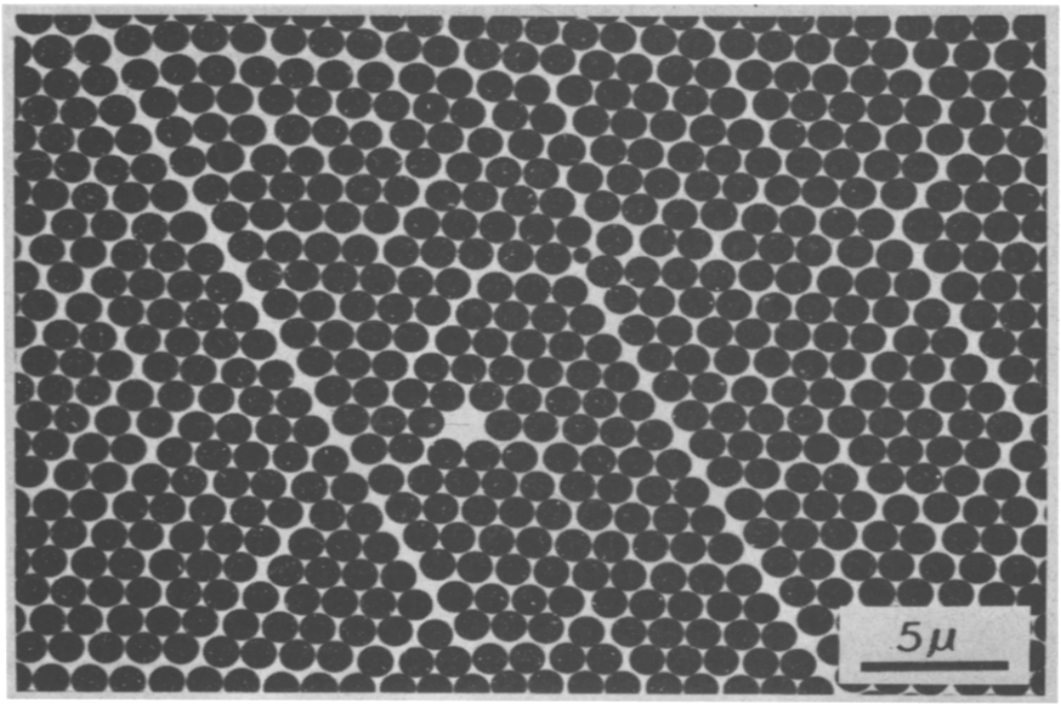


Fig. 1. Electron micrograph of a monodisperse polyvinyltoluene latex of diameter 1.24 μ, prepared by emulsion polymerization of styrene followed by four successive seeded overpolymerizations with vinyltoluene.

sections describe experimental and analytical approaches to the determination of the rheological equation of state, with the concluding section indicating some directions for future investigation.

HISTORICAL REVIEW

Einstein's inaugural dissertation at Zurich included the theory of the viscosity of a dilute dispersion of rigid spheres. With η the viscosity of the suspension, η_0 that of the medium, and ϕ the volume fraction of spheres in the suspension, Einstein's result for the relative viscosity $\eta_r = \eta/\eta_0$ can be written

$$\eta_r = 1 + 5\phi/2 + O(\phi^2) \tag{1}$$

where O means "terms of the order of". Thus the intrinsic viscosity $[\eta]$, defined as

$$[\eta] = \mathrm{Lim}_{\phi\to 0}(\eta_r - 1)/\phi \tag{2}$$

is precisely 5/2, regardless of the size or size distribution of the spheres. Experiments on systems ranging from neutrally buoyant glass sphere disper-

sions to polymer latices have generally confirmed Einstein's prediction, with measured intrinsic viscosities usually ranging from 2.50 to *ca.* 3.0.

The presence of electrical charges on the spheres causes an augmentation of the intrinsic viscosity, because of increased energy dissipation in the electrical double layer surrounding the particle. The magnitude of this electroviscous effect was first given without proof by von Smoluchowski[5], and recalculated by Krasny-Ergen[6], with a result identical except for a numerical factor. Subsequent analyses by Henry[7] and by Booth[8] indicated that the earlier estimates of the effect were too high. Booth's result is currently accepted as theoretically correct, although the recent experimental data of Chan and Goring[9] on sulfonated latex spheres fall between Booth's predictions and those of von Smoluchowski. They showed that the measured intrinsic viscosity is strongly dependent on the electrolyte concentration of the diluent, as is the case with polyelectrolytes.

At finite concentrations, the relative viscosity function

$$\eta_r = h(\phi) \tag{3}$$

behaves in accordance with Fig. 2. The dilute region is determined by the requirement that the slope at the origin equal the intrinsic viscosity, while

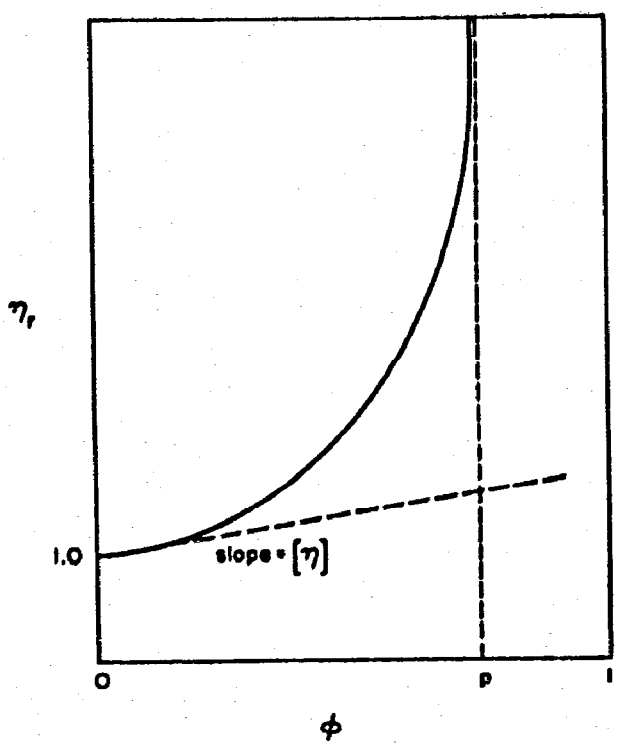


Fig. 2. Schematic graph showing the concentration dependence of the relative viscosity of a typical colloidal dispersion.

the behavior at high concentration is governed by the fact that beyond a volume fraction p, called the packing fraction, the dispersed particles lock into a rigid structure, and flow ceases. (A complication arises if non-Newtonian flow occurs at high concentrations; in such cases, the curve of Fig. 1 can be considered to represent a fixed value of either shear stress or shear rate.) The nature of the function $h(\phi)$ has been the subject of many studies, both experimental and theoretical. Eilers[10], in his study of suspensions of bitumen spheres, found that his data could be represented by the function

$$\eta_r = [1 + \tfrac{1}{2}[\eta]\phi/(1-\phi/p)]^2 \tag{4}$$

Maron et al.[11], who studied polydisperse synthetic latices, correlated their data by means of the relation

$$\eta_r = \exp[[\eta]\phi/(1-\phi/p)] \tag{5}$$

This expression was given a theoretical derivation by Mooney[12] using an interesting functional analysis. Considering the volume fraction of a suspension to be divided into two portions, he visualized the solution as being constituted by adding to a suspension of volume fraction ϕ_1 sufficient particles to raise the volume fraction to ϕ; or, alternatively, one could have started with a volume fraction $\phi_2 = \phi - \phi_1$. On the assumption that the starting suspension behaves in this process like a continuous medium of the same viscosity, the final suspension will have a relative viscosity given by

$$\eta_r(\phi_1+\phi_2) = \eta_r(\phi_1)\,\eta_r(\phi_2) \tag{6}$$

The solution of this functional equation is

$$\eta_r = \exp([\eta]\phi) \tag{7}$$

Although this solution behaves properly at low volume fractions, it does not increase sufficiently rapidly to represent real systems at high concentrations. Mooney modified this functional analysis by assuming that the volume available to disperse the second portion was diminished by a factor $1-K\phi$, due to the volume-plus-covolume pre-empted by the particles already present. Thus the effective volume fraction ϕ_{12} of the added particles in the presence of the first portion is

$$\phi_{12} = \phi_1/(1-K\phi_2) \tag{8}$$

An effective volume for the second fraction is defined as

$$\phi_{21} = \phi_2/(1-K\phi_1) \tag{9}$$

The functional equation of Mooney is thus

$$\eta_r(\phi_1+\phi_2) = \eta_r(\phi_{12})\,\eta_r(\phi_{21}) \tag{10}$$

whose solution is eqn. (5), with $p = 1/K$. Dougherty and Krieger[12a] contended that the crowding effect applies only to the second portion, although the relative viscosity must be independent of the order of addition; their functional equation is

$$\eta_r(\phi_1+\phi_2) = \eta_r(\phi_1)\eta_r(\phi_{21}) = \eta_r(\phi_2)\eta_r(\phi_{12}) \tag{11}$$

whose solution is

$$\eta_r = [1-(\phi/p)]^{-[\eta]p} \tag{12}$$

It can be shown that eqn. (12) becomes identical to eqn. (4) when $p[\eta] = 2$. Taking $[\eta] = 5/2$ and $p = 0.74$ (close-packed uniform spheres), the value of the product is 1.85; hence the equation of Dougherty and Krieger and that of Eilers predict very similar behavior for rigid-sphere dispersions.

Attempts to derive viscosity *vs.* concentration relationships on a more rigorous basis encounter difficult hydrodynamic problems, and have not as yet given results which accord better with experiment than those of the semi-continuum approach outlined above.

For the case of electrically charged particles, a "second electroviscous effect" augments the relative viscosity of concentrated suspensions. Fryling[13], and subsequently Brodnyan and Kelley[13a], dialytically removed all free electrolyte from concentrated latices, and thereby raised their viscosities by orders of magnitude. Brodnyan and Kelley attribute the effect to an augmentation of the effective hydrodynamic radius of the particle by the thickness of the electrical double layer, an explanation employed also by Wang[14] to explain his observations on the effect of surface charge and electrolyte content on the viscosities of monodisperse latices. Although this explanation accords qualitatively with their data, it can not be rigorously valid, since most of the double layer does not rotate with the particle. The explanation of Harmsen et al.[15], in terms of displacements of the paths of neighboring particles due to electrostatic repulsions, is more reasonable, and likewise depends on double-layer thickness. It also accords with the vanishing of the second electroviscous effect at high shear rates, which is discussed below. The calculation of Chan et al.[16], based on a collision doublet model, overestimates the second electroviscous effect, according to the measurements of Stone-Masui and Watillon[17].

Early studies of non-Newtonian behavior in latices postulated Bingham-body behavior, with no flow at shear stresses below the "yield stress". Extensive experiments by Maron et al.[18,19] disproved the existence of a yield point, and showed that non-Newtonian flow in commercial latices occurs only at volume fractions above *ca.* 0.25. They used both the "power-law" and the Ree–Eyring equation[20] with two or three flow units to express the dependence of viscosity on the shear rate $\dot{\gamma}$. The power law can be written in

terms of the viscosity $\eta_{(1)}$ at unit shear rate $\dot{\gamma}_{(1)}$

$$\eta = \eta_{(1)}(\dot{\gamma}/\dot{\gamma}_{(1)})^m \tag{13}$$

where m is an index of non-Newtonian behavior; negative m indicates shear-thinning and positive m shear-thickening, while for Newtonian behavior $m = 0$. The Ree–Eyring equation involves a viscosity parameter a_j and a relaxation time λ_j for each of $n+1$ flow units, with $\lambda_0 = 0$.

$$\eta = a_0 + \sum_{j=1}^{n} a_j(\lambda_j\dot{\gamma})^{-1} \sinh^{-1}(\lambda_j\dot{\gamma}) \tag{14}$$

Maron and Madow[21] also investigated the effect of polydispersity. Mixing two latices of equal volume fractions and different average particle diameters, they found that the mixture was more fluid and more nearly Newtonian than either of the parent latices.

Krieger and Dougherty[22] derived an equation for non-Newtonian viscosity of rigid sphere dispersions, in which nearby particle pairs rotate as doublets in a shear field; these proximity doublets are formed by Brownian movement and destroyed by shear. They were able to express their result in the form of the Williamson equation[23]

$$\eta = \eta_1 + (\eta_2 - \eta_1)(1 + |\tau|/\tau_m)^{-1} \tag{15}$$

in which the viscosity decreases with shear from a zero-shear limit η_2 to a high-shear limit η_1. The parameter τ_m, which is physically the shear stress at which the viscosity lies midway between η_1 and η_2, is concentration-independent

$$\tau_m = \alpha kT/a^3 \tag{16}$$

where a is the particle radius, k Boltzmann's constant, T the temperature and α an incompletely determined number of order 1. The equation was found to fit the latex data of Maron and Fok[24] and Maron and Levy-Pascal[25] at all concentrations below 55% by volume; above this concentration it became necessary to vary τ_m with concentration to obtain a good fit.

Little work has been published on time-dependent thixotropic effects in latices. A dimensional analysis of the rheological behavior of rigid sphere dispersions, discussed in a later section of this paper, yields a time constant $t^* = \eta_0 a^3/kT$. For a typical latex, t^* is of the order of 5 msec, indicating that time-dependent effects are of short duration compared with the response times of most rheological instruments. A possibly relevant observation is that of Mooney[26], who studied a thickened natural latex. He found that when the shear rate was suddenly reduced to a new steady value, the viscosity increased with time, approaching asymptotically the steady-state viscosity at

the new shear rate. The particle size in natural latex is several times as large as in a typical synthetic latex, and the presence of a water-soluble thickener increases η_0; apparently the combination of the two effects increased t^* to a value readily detectable with conventional instrumentation.

With the discovery after 1950 of methods to produce monodisperse latices over a wide range of sizes, control of the rheologically important variable of particle size distribution became feasible. Saunders[27] studied the viscosities of monodisperse polystyrene latices with diameters ranging from 0.1 to 0.9 μ. In order to control the surfactant level, he added surfactant until the surface tension dropped to the level characteristic of surface saturation, and performed all dilutions with a surfactant solution which also represented the surface tension minimum. By including in the sphere's diameter an increment for the surfactant monolayer, Saunders found an intrinsic viscosity of 2.504 ± 0.046. At higher concentrations, Mooney's equation represented the data well, but the value of p varied with particle diameter. Schaller and Humphrey[28] studied monodisperse latices of 0.088, 0.557, and 1.305 μ diameter at volume fractions up to 25%. They found Newtonian behavior in all cases except for the smallest particles at low electrolyte level, and also utilized Mooney's equation. Woods and Krieger[29] showed that a concentrated monodisperse polymer latex behaves like the model hard-sphere suspension under the following conditions: (*1*) particle diameters below 0.5 μ; (2) presence of sufficient neutral electrolyte to provide a compact double layer which screens interparticle Coulombic forces; and (*3*) inclusion of the surfactant monolayer thickness as part of the particle radius.

An investigation by Dodge and Krieger[30] of the laminar–turbulent transition of monodisperse latices in tube flow showed no gross anomalies, although some suppression of the transition was found. In most instances, critical Reynolds numbers close to the "normal" value of 2000 were observed. With large particles (diameters in excess of 0.5 μ) and at high concentrations, however, critical Reynolds numbers exceeded 5000. A curious effect due to the surfactant was noted, in that latices prepared with high proportions of anionic surfactants consistently showed critical Reynolds numbers near 3000; the suppressive effect was confined to latices with predominantly nonionic surfactant.

COLLOID CHEMISTRY OF SYNTHETIC LATICES

In the emulsion polymerization process, water, surfactant, vinyl monomer and a free radical initiator are charged into a stirred reactor. For the case of slightly water-soluble monomers such as styrene, and water-soluble initiators such as potassium persulfate, the free radicals are believed to attack monomer molecules in the aqueous phase[31]; the resulting radicals, which may be aug-

mented by addition of monomer molecules, are eventually captured by monomer-swollen surfactant micelles. Here the radical grows by chain addition until terminated by reaction when a second radical is captured by the same micelle. Thus the polymerization reaction within a micelle is alternatively switched on and off as the total number of radicals captured changes from odd to even. Monodispersity is usually achieved by limiting the number of micelles, so that all potential polymerization sites are activated early in the reaction cycle, and the growth time for all particles is approximately the same.

After the reaction has gone essentially to competition, and any residual monomer has been distilled away, the resulting latex is an aqueous dispersion of spherical polymer particles, with surfactant molecules or radicals adsorbed on their surfaces. A single sphere consists of many polymer molecules, each terminated at both ends by an initiator fragment. Generally, the initiator radical is either ionic or highly polar, and therefore its preferred location is at the water–polymer interface. For hydrogen peroxide-initiated latices, the surface hydroxyl groups are usually oxidized to carboxyl groups, which are weakly acidic. When persulfate is used as initiator, most of the chain terminations are the strongly acidic sulfate groups; however, hydroxyl radicals may be generated by the reaction

$$\cdot SO_4^- + HOH \rightarrow HSO_4^- + \cdot OH \qquad (17)$$

so that even in this case an appreciable fraction of the chain ends may be carboxyl groups. Through control of pH during persulfate-initiated polymerizations, the number of carboxyl groups can be greatly reduced. The aqueous medium contains unadsorbed surfactant, counterions to the bound or adsorbed charges, and free electrolyte resulting from the initiator or from impurities accompanying the surfactant.

Early rheological studies were performed on "as polymerized" latices, which were usually polydisperse. Techniques now available permit the synthesis of highly monodisperse latices, and the subsequent removal of all soluble impurities. Dialysis against distilled water removes all ions with the exception of the bound charges on the particles and sufficient counterions to achieve electroneutrality. Deionization can be achieved more easily using ion exchange techniques[32], but nonionic surfactants, if present, require dialysis for their removal. A recent article[33] describes a method for controllably increasing particle charge through *in situ* polymerization of surface-active monomer on the particle surface.

A type of dispersion closely related to latices consists of a non-aqueous dispersion (NAD) of polymer spheres. NAD's can be formed either by direct polymerization, typically of acrylic monomers in a hydrocarbon medium, or by redispersal of latex spheres. The direct polymerization methods utilize relatively short-chain molecules bound or sorbed at the polymer surface;

these stabilize the dispersion by an effect designated as "steric" or "osmotic", which is similar or identical to the mechanism by which non-ionic surfactants stabilize aqueous latices. Although there are no published descriptions of techniques for producing monodisperse NAD's by direct polymerization, they have been produced in the author's laboratory by redispersal of cross-linked monodisperse latex spheres. This technique involves ion exchange, followed by two successive dialyses, first against methanol and then against the final medium. The most stable re-dispersions have been effected in somewhat polar media, where it has been shown electrophoretically that the particles are negatively charged. With such techniques, the experimentalist can now control charge, size and size distribution of the particles, type and concentration of surfactant, and the electrical properties of the aqueous phase. The fact that these capabilities have not as yet been fully utilized in a rheological investigation is due solely to their newness.

Direct determination of particle size and size distribution is achievable by electron microscopy. Letting n_i be the number of particles whose measured diameter is D_i, the mth moment $\langle D^m \rangle$ is defined as

$$\langle D^m \rangle = \Sigma n_i D_i^m / \Sigma n_i \tag{18}$$

Various average particle sizes frequently quoted are the number average $D_n = \langle D \rangle$, the weight average given by $D_w^3 = \langle D^6 \rangle / \langle D^3 \rangle$, and the volume-to-surface average $D_s = \langle D^3 \rangle / \langle D^2 \rangle$. Indices of polydispersity are the standard deviation σ_D, given by $\sigma_D^2 = \langle D^2 \rangle - \langle D \rangle^2$, and the uniformity index $U = D_w / D_n$. Uniformity indices below 1.01 are frequently achieved, corresponding to relative standard deviations σ_D / D_n of less than 5%. For these narrow distributions, the mth moment is conveniently estimated by the relation[34]

$$\langle D^m \rangle / \langle D \rangle^m = (1 + S^2)^{m(m-1)/2} \tag{19}$$

where $S = \sigma_D / \langle D \rangle$.

In order to determine bound surface charge, the latex is first deionized, so that ionic species are removed and the surface charges are converted to their protonated forms. These are titrated conductimetrically against standard base; strong and weak acid groups can be separately determined from the titration curves. The degree of surfactant coverage of the particle surface is found by following the course of a titration with surfactant by means of surface tension measurements; an abrupt change to a horizontal slope occurs at the point of surface saturation.

A PRINCIPLE OF CORRESPONDING RHEOLOGICAL STATES

A rheological equation of state is a relation giving the rheological properties of a fluid in terms of shear variables and of those structural parameters

which determine its mechanical response. For a colloidal dispersion of hard spheres, the rheological property of interest is the viscosity η of the suspension, and the shear variables are the shear rate $\dot{\gamma}$ (or, alternatively, the shear stress τ) and the time t. Since the spheres are much larger than the molecules of the medium, only continuum properties of the medium are of interest: its viscosity η_0 and its density ρ_0. The radius, a, of the particles, their concentration (number density) n, and their density ρ_p are relevant, and, whenever Brownian movement is significant, a thermal energy variable kT, where k is Boltzmann's constant and T the temperature. The equation of state can be written[35]

$$\eta = f_1(\dot{\gamma}, t, \eta_0, \rho_0, a, n, \rho_p, kT) \tag{20}$$

In this 9-variable equation, all of the variables are expressible in terms of the 3 basic units of mass, length and time; hence, according to the principles of dimensional analysis, the equation can be formulated in terms of $9-3=6$ dimensionless groups. A convenient set of 6 independent groups consists of relative viscosity $\eta_r = \eta/\eta_0$, the volume fraction $\phi = 4\pi n a^3/3$, a reduced shear stress $\tau_r = \tau a^3/(kT)$, a reduced time $t_r = t/t^*$ (where $t^* = \eta_0 a^3/(kT)$), a relative density $\rho_r = \rho_p/\rho_0$, and an internal Reynolds number $R_i = a^2\dot{\gamma}\rho_0/\eta_0$. The equation of state in reduced form is

$$\eta_r = f(\phi, \tau_r, t_r, \rho_r, R_i) \tag{21}$$

The form of the function f must be found either empirically or through theoretical analysis of the flow process. The task is simplified by initially restricting consideration to steady ($t_r \to \infty$) shear of a neutrally buoyant ($\rho_r \to 1$) suspension under laminar flow conditions ($R_i \to 0$); in this limit eqn. (21) become

$$\eta_r = f(\phi, \tau_r, \infty, 1, 0) = f(\phi, \tau_r) \tag{22}$$

This "principle of corresponding rheological states" requires that *two hard-sphere dispersions at the same volume fraction and reduced stress exhibit the same relative viscosity*, even if they differ greatly in particle size or in viscosity of the medium.

Lifting the restriction of steady shear re-introduces the reduced time variable

$$\eta_r = f(\phi, \tau_r, t_r) \tag{23}$$

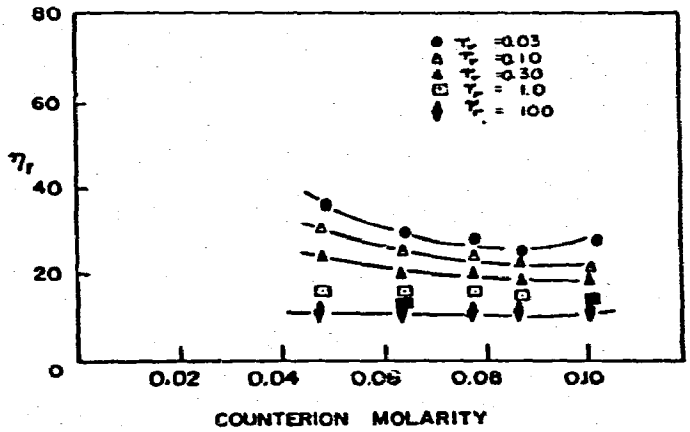
This is a corresponding-states principle for systems at the same volume fraction which are subjected to the same mechanical history $\tau_r(t_r)$. Thus, for example, if two hard-sphere dispersions at the same volume fraction are brought instantaneously from a stress-free state to the same reduced shear stress τ_r, then the graphs of their relative viscosities *vs.* reduced time should superpose.

The restriction imposed by the hard-sphere model must be borne in mind in applying the principles developed above to latex systems. Not only must the particles be uniform, spherical and rigid, but their interactions must be limited to hard-sphere repulsions and mechanical stresses propagated through the medium. Thus any Coulombic or van der Waals forces must be negligible in comparison to shearing and Brownian forces.

THE RHEOLOGICAL EQUATION OF STATE: EXPERIMENTAL

This section describes an experimental program carried out in the author's laboratory with the objective of determining empirically for hard-sphere dispersions the functional form of the rheological equation of state $\eta_r = f(\phi, \tau_r)$. Monodisperse polystyrene and polyvinyltoluene latices were prepared by special emulsion polymerization techniques[4] using mixtures of anionic and nonionic surfactants, resulting in diameters ranging from 0.19 to 1.12 μ, with relative standard deviations less than 4%. Non-Newtonian viscosities were measured as functions of volume fraction, surfactant content and electrolyte level, using capillary, cone-plate, Couette and parallel-disk viscometers[36].

The effect of electrolyte level is shown in Figs. 3 and 4, where relative viscosities of two different latices are graphed *vs.* electrolyte content at various values of τ_r. At sufficiently high shear stresses, the viscosity is independent o electrolyte content, but at lower stresses a distinct minimum is observed; the corresponding electrolyte content c_{min} is independent of shear stress.



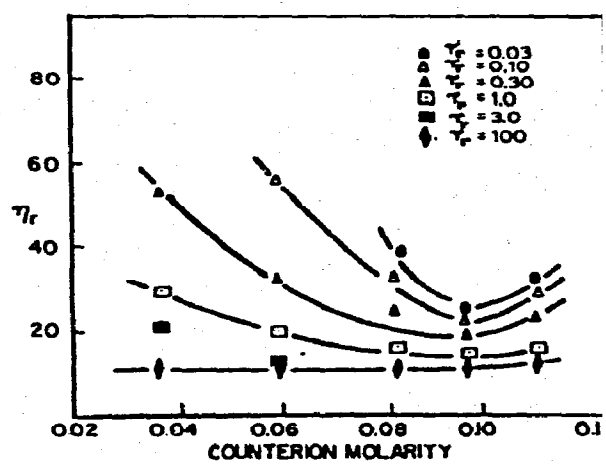


Fig. 3. Graph showing the effect of electrolyte on the non-Newtonian viscosity of monodisperse polystyrene latex: $D_0 = 0.192\mu$, $\phi = 50.9\%$, pH = 2.7; surfactant is 2% anionic, 98% nonionic.

Fig. 4. Graph showing the effect of electrolyte on the non-Newtonian viscosity of the same latex as in Fig. 3, but with higher content of anionic surfactant: $D_0 = 0.192\mu$, $\phi = 50.9\%$, pH = 7.0; surfactant is 21% anionic, 79% nonionic.

The minimum is more pronounced for the latex containing the higher amount of anionic surfactant. Latices containing more electrolyte than c_{min} frequently showed visible graininess, indicating the presence of agglomerates. Since any inter-particle Coulombic or van der Waals forces would increase the viscosity above the hard-sphere limit, c_{min} was taken as the electrolyte

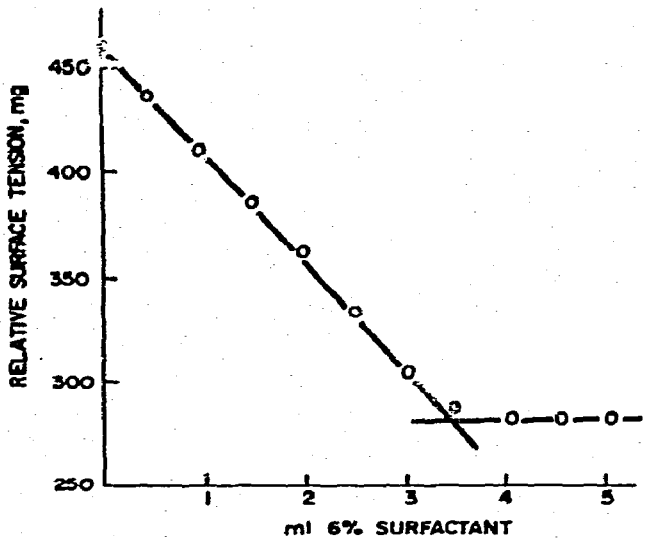


Fig. 5. Typical graph showing variation of surface tension when a latex is titrated with surfactant. The break is taken to indicate the point of 100% surface coverage.

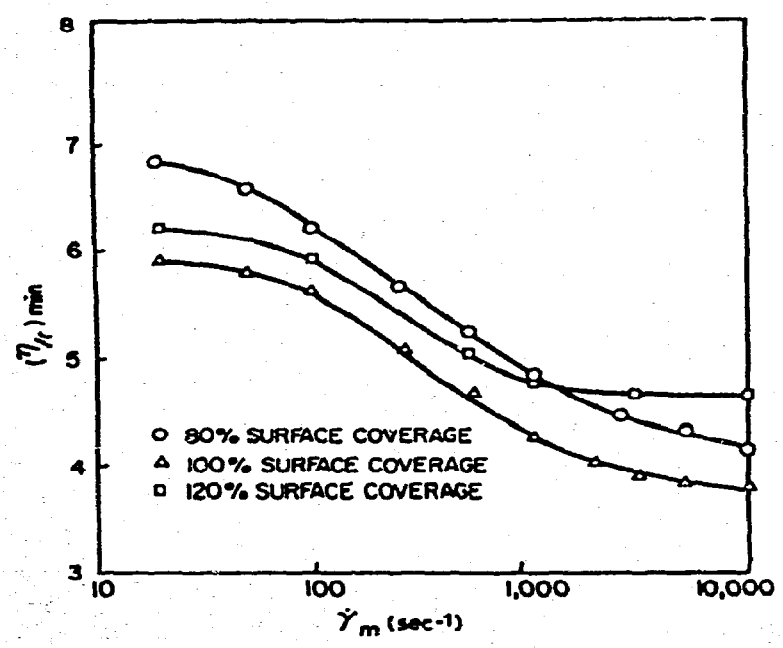


Fig. 6. Effect of surfactant coverage on the non-Newtonian viscosity of a monodisperse latex. $D_0 = 0.21\,\mu$, $\phi = 42.4\%$.

level which provides maximum shielding of Coulombic forces without destabilizing the latex. All rheological measurements on aqueous latices described below were preceded by addition of the quantity of electrolyte required to attain the viscosity minimum.

The surfactant level corresponding to 100% monolayer coverage of the particle surface was determined by surface tension measurements, as shown in Fig. 5. Latices were prepared with 80%, 100%, and 120% surface saturation; as shown in Fig. 6, the sample at 100% coverage exhibited the lowest viscosity. The increase in viscosity beyond 100% coverage is probably due to micellization of the excess surfactant; the increase below 100% coverage may be due to a tendency toward aggregation. Data reported below were taken on latices whose surface coverage was adjusted to slightly below 100% by addition of anionic surfactant.

In order to take into account the increase of volume fraction of the disperse phase due to the adsorbed surfactant monolayer, use was made of the empirical fact that the function $f(\phi,\tau_r)$ approaches a limit η_{1r} as τ_r increases.

$$\mathrm{Lim}_{\tau_r\to\infty}\ \eta_r = \eta_{1r} \tag{24}$$

These limiting viscosities were measured at various polymer volume fractions ϕ_p, for particle diameters of 0.20, 0.66 and 1.12 μ, with results presented as open points on Fig.6(a). Utilizing a value $\Delta \approx 45$ Å, given by Kushner and Hubbard[37] as the fully extended length of the nonionic surfactant (Triton X-100, Rohm and Haas Co.) which constituted *ca.* 80% of the surfactant present in the latex, an effective volume fraction $\phi = \phi_p(1 + 6\Delta/D)$ was computed. When η_{1r} was plotted against ϕ, the closed points of Fig. 6(a) were obtained. It is evident that this correction places the points for the different particle sizes onto a single curve. This method of confirming the surfactant layer thickness makes use of data at high volume fraction, in contrast to the method used by Saunders and by Selby and Heller[38], which uses only the intrinsic viscosity. Volume fractions reported below incorporate this correction.

Viscosity *vs.* shear stress measurements were made on latices of diameters 0.217, 0.280, 0.360, 0.513 and 0.628 μ at volume fractions up to 55%. Results for $\phi = 0.50$ are presented in Fig. 7. As may be seen, superposition is excellent for the three smaller sizes, but the corresponding-states principle is not obeyed by the larger particles. There is independent evidence from sedimentation behavior that large particles form long-lived aggregates, due presumably to the secondary minimum in the interparticle potential curve which arises from van der Waals forces. (For example, a polystyrene latex of diameter 0.4 μ shows no settling on prolonged storage, whereas a 0.63 μ polyvinyltoluene latex settles appreciably. Since polystyrene has a density of 1.05 g/cc and polyvinyltoluene 1.02 g/cc, the sedimentation rates should be virtually identical.) Interestingly, relative viscosities for the two large sizes superposed

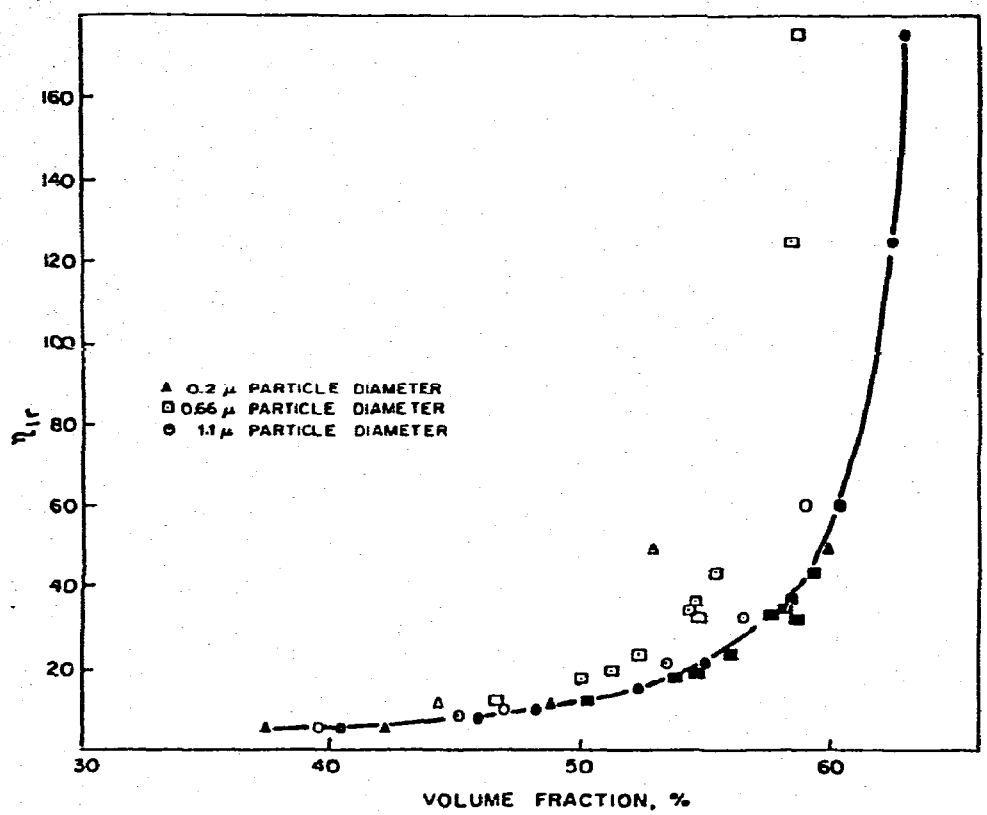


Fig. 6(a). High-shear limiting viscosity *vs.* polymer volume fraction (open symbols) and *vs.* effective volume fraction (closed symbols).

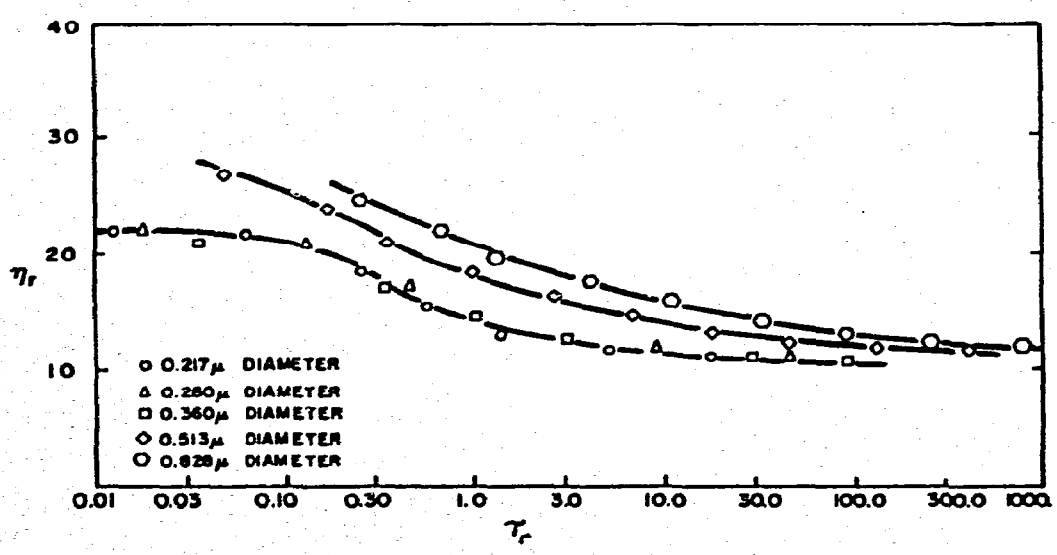


Fig. 7. Relative viscosity *vs.* reduced shear stress for aqueous monodisperse latices at 5 different particle sizes: ϕ = 50.0%. Note superposition of three lowest particle sizes.

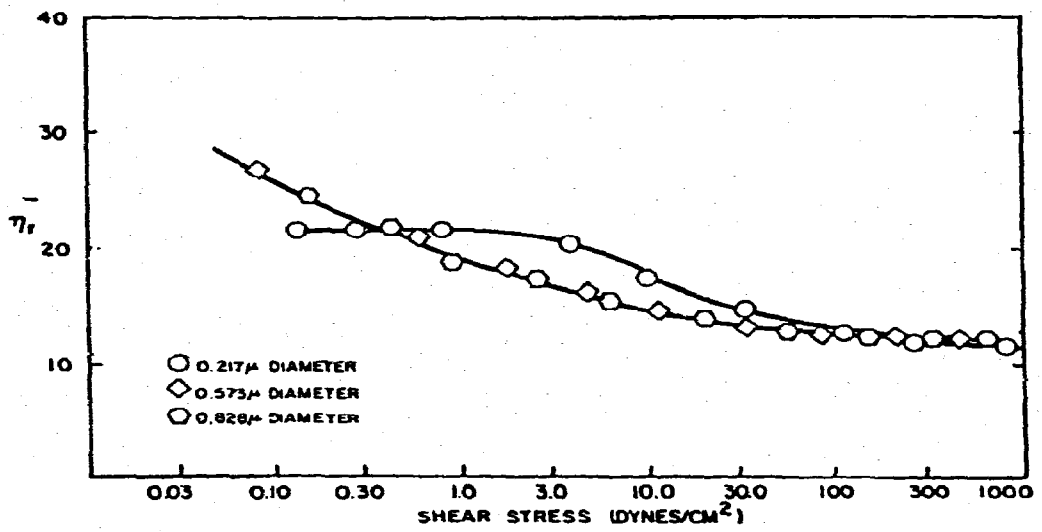


Fig. 8. Data of Fig. 7 graphed against actual shear stress. Points for 0.280μ and 0.360μ latices were omitted for clarity.

when graphed against the shear stress τ (Fig. 8), while the three smaller sizes gave separate curves. (To avoid confusion, points for the 0.280 and 0.360μ latices were omitted from the graph.)

From the evidence cited above, it may be inferred that aqueous latices whose diameters are below 0.5 μ will behave as hard-sphere dispersions if sufficient electrolyte is present to provide effective screening of interparticle Coulombic forces. There remains, however, the question of whether the electroviscous effect has indeed been rendered negligible by this means. In order to answer this question, an alternative methods of suppressing Coulombic forces was explored. By resuspending the polymer spheres in less polar media, the degree of electrolytic dissociation of the surface groups is substantially reduced. Resuspension has the additional advantage that it permits varying the viscosity η_0 of the medium. (Variation of η_0 could also be achieved by adding water-soluble thickeners to aqueous latices, but this would give rise to new problems as a result of adsorption of solute on the polymer spheres.) Accordingly, techniques were developed to resuspend latex spheres in suitable organic liquids, and to measure the viscosities of the resulting non-aqueous dispersions[39].

Since linear polystyrene is readily swelled or dissolved by most organic liquids, the first step in the program was to prepare crosslinked monodisperse latices. Emulsion polymerization recipes were developed which permit replacement of up to 10 mole % of the styrene monomer by the crosslinking agent divinylbenzene. Diameters achieved ranged from 0.15 μ to 0.43 μ, with standard deviations of *ca.* 5%. The technique described below for resuspend-

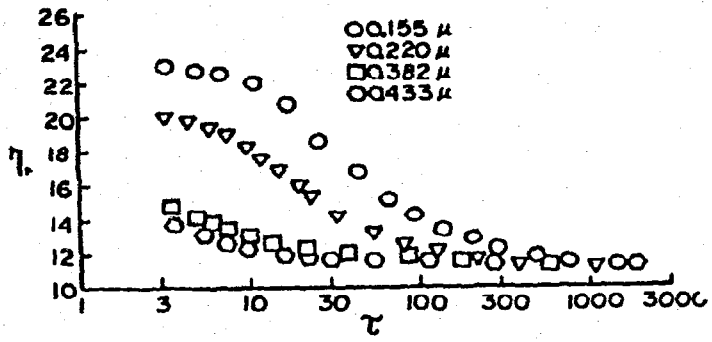


Fig. 9. Relative viscosities *vs.* shear stress for 50% monodispersions in benzyl alcohol of four different particle sizes of crosslinked polystyrene spheres.

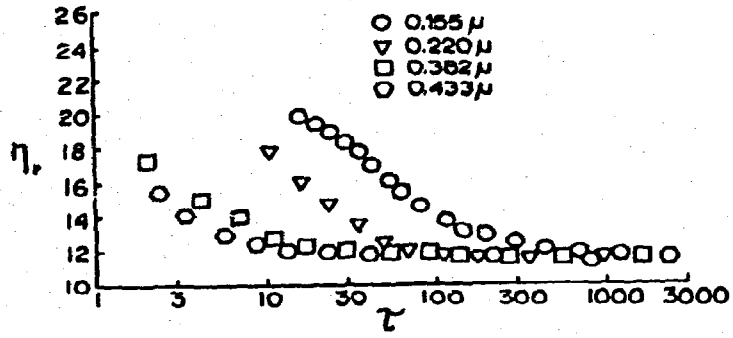


Fig. 10. Relative viscosities *vs.* shear stress for 50% monodispersions in *m*-cresol of four different particle sizes of crosslinked polystyrene spheres.

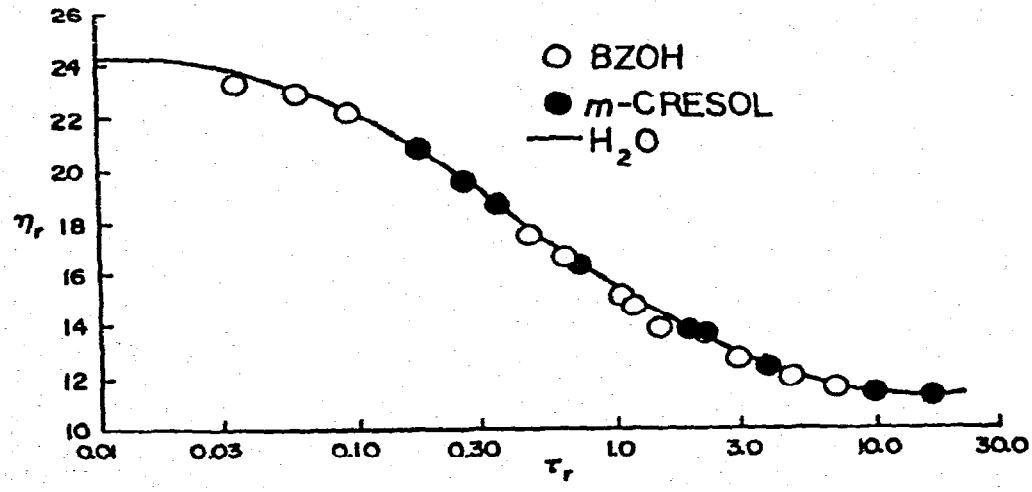


Fig. 11. Relative viscosities *vs.* reduced shear stress for 50% monodispersions of polystyrene spheres of various sizes in different media. Points are taken from Figs. 9 and 10; solid line from aqueous data of Fig. 7.

ing the crosslinked polymer spheres, although elaborate and time-consuming, is reproducibly applicable to a wide variety of organic media.

The crosslinked latex is first deionized using a monobed ion exchange resin. The second step is dialysis against methanol, chosen as an intermediate medium because of its miscibility with both water and organic liquids. The methanol suspension, at a polymer concentration of 15–20% by volume, is next dialyzed against the desired final medium. Regenerated cellulose tubing was used as the dialytic membrane. A rotary evaporator was used to remove undesired medium. When the final medium is miscible with water, the intermediate dialysis against methanol may be omitted.

TABLE 1

Intrinsic viscosities of suspensions of monodisperse crosslinked polystyrene in various media

Medium	$[\eta]_\phi$
Benzyl alcohol	2.67
m-Cresol	2.68
Benzene	5.84
Toluene	7.50
Carbon tetrachloride	6.89

Stable resuspensions were achieved by this technique in a wide range of organic media. Resuspensions in moderately polar media such as benzyl alcohol and *m*-cresol exhibited the iridescence characteristic of monodisperse aqueous latices, and intrinsic viscosities near the Einstein value of 5/2 (see Table 1). Non-polar liquids such as benzene, toluene and CCl_4 gave colorless resuspensions whose intrinsic viscosities were higher by factors of 2 to 3. The high values were attributed to solvent attack on the polymer spheres, resulting in swelling and partial dissolution. Benzyl alcohol and *m*-cresol were selected as suitable non-swelling media; their viscosities at 30°C are 0.0489 poise and 0.1738 poise respectively. Suspensions of each of four particle sizes (0.155 μ, 0.220 μ, 0.382 μ and 0.433 μ) were prepared in both media at volume fractions of 0.50.

Non-Newtonian viscosity measurements of all eight systems were made at volume fractions of 0.30, 0.35, 0.40, 0.45 and 0.50. Figures 9 and 10 show the results at $\phi = 0.50$, plotted as η_0 *vs.* the unreduced shear stress τ. Figure 11 shows the same data graphed as η_0 *vs.* the reduced shear stress $\tau_r = a^3\tau/(kT)$. The solid line on Fig. 11 is taken from Fig. 7 and therefore represents the aqueous latices at the same volume fraction. The excellent concordance, covering 20-fold ranges of both a^3 and η_0 (and hence a 400-fold range of t^*), confirms both the hard-sphere model and the corresponding-states principle.

THE RHEOLOGICAL EQUATION OF STATE: THEORETICAL

A theory has been developed for non-Newtonian flow in hard-sphere dispersions[22]. Although primitive and inexact, it furnishes insight into more realistic mechanisms, and its results correspond reasonably well to the experimental data of the previous section.

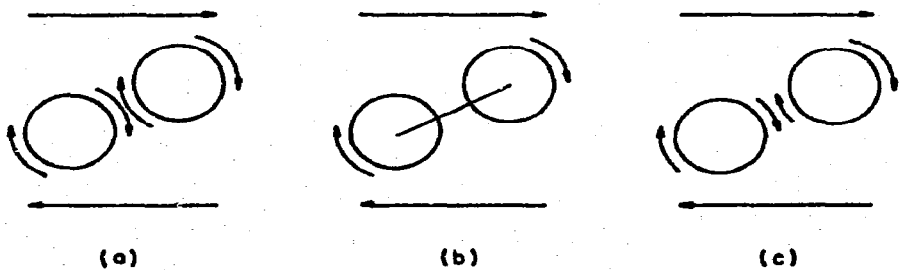


Fig. 12. Schematic graph showing how proximity doublets, formed by Brownian diffusion, are destroyed by shear. (a) Fluid motions if particles were to rotate independently; (b) rigid-dumbbell rotation; (c) a more realistic flow pattern, intermediate between (a) and (b).

Consider first a liquid under a uniform steady rate $\dot{\gamma}$. When an isolated rigid sphere is present, the fluid velocity in the neighborhood of the sphere is perturbed, and the sphere itself rotates with a steady angular velocity $\omega = \dot{\gamma}/2$. If, now, a second sphere is introduced within the region of perturbation, the two spheres can no longer rotate independently; rather, the perturbations in the velocity of the medium about sphere 1 give rise to a stress on sphere 2, and conversely. As a result, the pair of spheres will take on a doublet rotation about their mutual center of mass. The doublet rotation brings the spheres into fluid lamina of different overall velocities, and hence the particles will separate. Steps in the process are indicated schematically in Fig. 12. The essential features are that doublet motion arises from purely hydrodynamic forces between nearby particles, and that these proximity doublets, which in the static fluid are created and destroyed by Brownian movement, are destroyed by doublet rotation. Since a rotating doublet causes more dissipation of energy than would two isolated spheres, shear-induced destruction of doublets causes the system to behave as a shear-thinning non-Newtonian fluid.

Certain crude approximations permit the elaboration of this mechanism into a quantitative theory. First is an oversimplification which considers the spheres to be completely independent unless their centers are within some critical multiple of a diameter, in which case the pair is considered as a rigid doublet. The doublet concentration is assumed to be small, and higher aggregates are neglected entirely. Further assumptions are made regarding the specific rate constants for doublet break-up by Brownian diffusion and by shear-

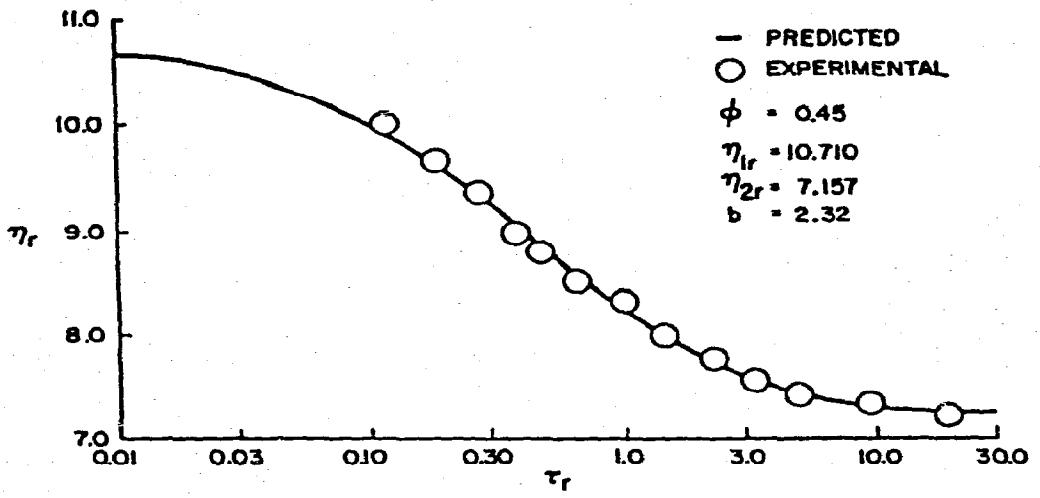


Fig. 13. Test of Williamson-type equation for 45% aqueous and non-aqueous monodispersions.

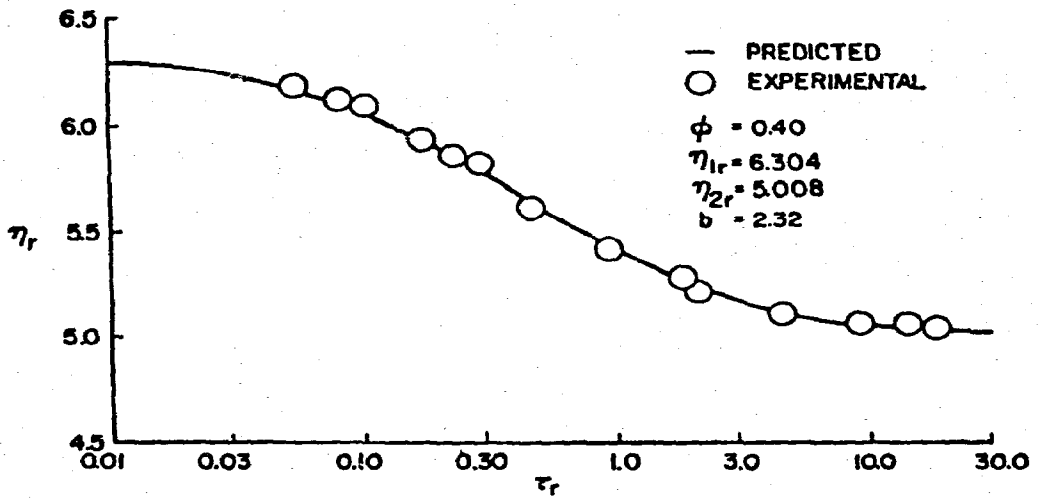


Fig. 14. Test of Williamson-type equation for 40% aqueous and non-aqueous monodispersions.

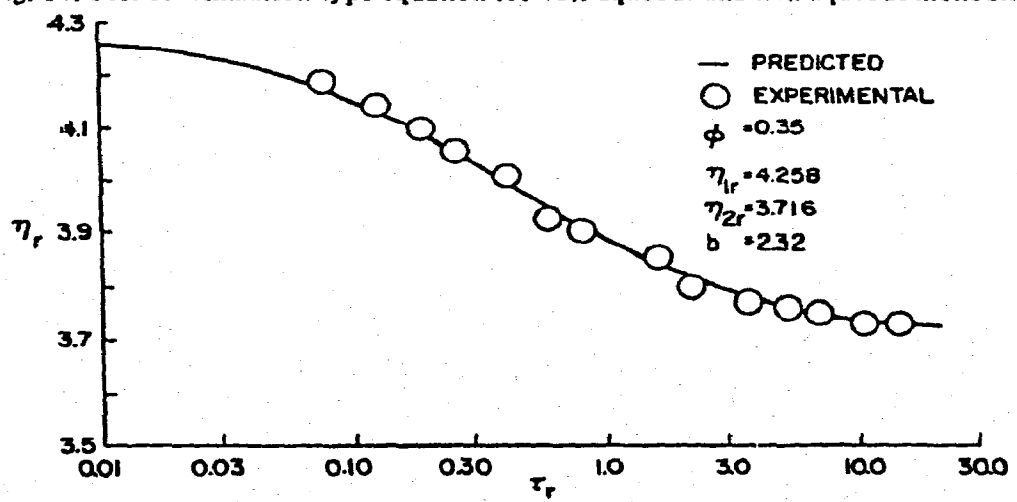


Fig. 15. Test of Williamson-type equation for 35% aqueous and non-aqueous monodispersions.

induced rotation. The resultant theory predicts that the relative viscosity will decrease from a zero-shear value η_{2r} to a high-shear limit η_{1r} according to a Williamson-type equation

$$(\eta_r - \eta_{1r})/(\eta_{2r} - \eta_{1r}) = (1 + |\tau|/\tau_c)^{-1} \tag{25}$$

where

$$\tau_c = kT/(3\alpha a^3) \tag{26}$$

Here α is a number of order unity whose value must be determined experimentally.

Equation (25) can be tested by fitting it to experimental data on hard-sphere dispersions. The theory requires, in addition to a "good fit", that τ_c-values be independent of concentration, and that they vary as the inverse third power of the particle diameter. When the equation was fitted by a least-squares method to the data of Fig. 11, representing hard-sphere dispersions at $\phi = 0.50$, the fitting parameters were found to be $\eta_{1r} = 24.7$, $\eta_{2r} = 10.8$ and $b = 2.32$. Using the same b value, the data for volume fractions of 0.45, 0.40 and 0.35 were fitted; the results are shown in Figs. 13, 14 and 15, and in Table 2.

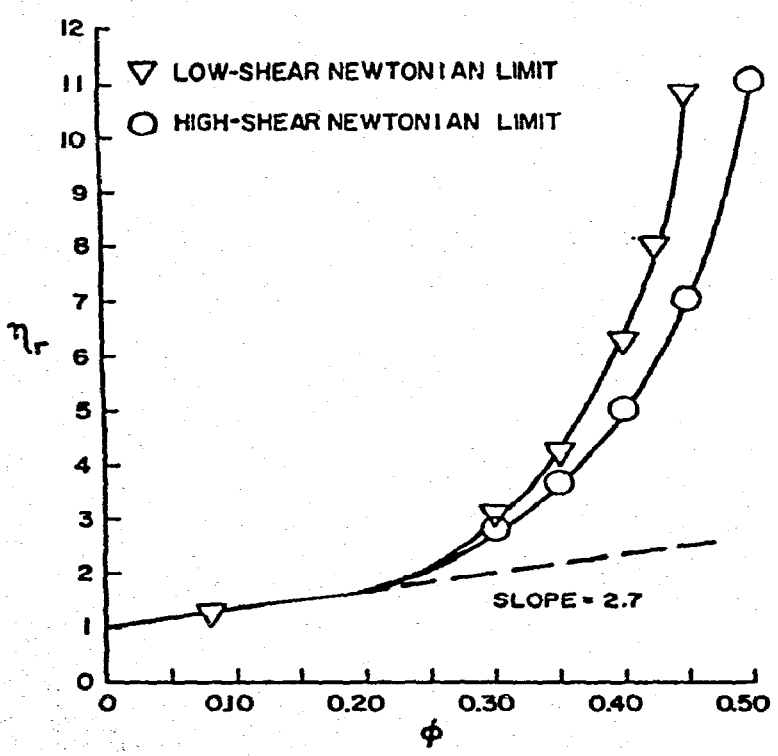


Fig. 16. Test of Dougherty–Krieger equation for concentration dependence of high- and low-shear limiting viscosities. Points are taken from Table 2; lines represents eqns. (27) and (28).

TABLE 2

Parameters obtained by fitting Williamson equation to the data of Figs. 11, 13, 14 and 15

ϕ (%)	b	η_{1r}	η_{2r}
50.0	2.32	10.8	24.7
45.0	2.32	7.157	10.710
40.0	2.32	5.008	6.304
35.0	2.32	3.716	4.258

In order to complete the analytical characterization of $\eta_r = f(\phi, \tau_r)$, it is necessary to express η_{1r} and η_{2r} as functions of ϕ. Mooney's equation, (5), was found to fit poorly, but both Eilers' equation, eqs. (4), and the Dougherty–Krieger equation, eqn. (12), proved suitable. Figure 16 shows the data of Table 2, together with the fitted curve of eqn. (12). Here $[\eta] = 2.67$ was used for both curves, with p values of 0.68 and 0.57 for the respective curves representing η_{1r} and η_{2r}. Thus the rheological equation of state for rigid-sphere dispersions in steady flow can be expressed in parametric form as

$$\eta_r = \eta_{1r} + (\eta_{2r} - \eta_{1r})/(1 + 0.431\,|\tau_r|) \tag{27}$$

with

$$\eta_{1r} = (1 - \phi/0.68)^{-1.82} \tag{28}$$

$$\eta_{2r} = (1 - \phi/0.57)^{-1.50} \tag{29}$$

$$\tau_r = \tau a^3/(kT) \tag{30}$$

These expressions permit the prediction of the steady-state viscosity of a uniform hard-sphere suspension of any diameter in a medium of any viscosity, at any volume fraction or shear stress. Significant discrepancies between prediction and observation may thus be attributed to differences between model and experimental systems.

ELECTROVISCOUS EFFECTS

According to the classification scheme of Conway and Dobry-Duclaux[40], there exist three distinguishable electroviscous effects: (*1*) an increment to the intrinsic viscosity due to distortion under shear of the electrical double layer; (*2*) an increase in the viscosity at finite concentrations due to Coulombic interactions between the double layers of different particles; and (*3*) effects due to the distortion of the particle itself as a result of electrostatic forces arising from the presence of charges on the particle. The total absence of the third electroviscous effect in rigid-particle systems makes them ideally suitable for study of the first two effects.

Accurate measurements of intrinsic viscosities are not easily performed, since they depend upon extrapolation of small differences. The primary electroviscous effect in monodisperse latices produces only a small increment in the intrinsic viscosity over the Einstein value for uncharged spheres. Other increments, such as those due to non-sphericity or to adsorption of surface-active molecules, can mask the electroviscous contribution. A successful investigation therefore requires fully characterized systems, well-designed and meticulously performed measurements, and careful interpretation of the results. The recent advances in emulsion polymerization, surface treatment and purification of monodisperse latices make available a wide range of particle sizes and surface charges. Definitive measurements of the primary electroviscous effect should now be within reach.

The second electroviscous effect can alter the viscosity of a monodisperse latex by an order of magnitude, and hence its measurement makes lesser demands on the experimenter. The effect is most pronounced in deionized latices; however, under these conditions the viscosity is strongly affected by traces of electrolytic impurities. As in the case of the primary effect, recent advances make possible definitive experiments.

For a system of charged particles, additional structural variables must be present in the equation of state. These include the particle charge q, the added ionic concentration c (assuming a $1-1$ electrolyte), and the dielectric constant D of the medium. The electronic charge ϵ must also appear. Four new variables are thus added to the equation, each expressible in terms of mass, length and time, and hence there will be four new independent dimensionless groups. A suitable set is D, q/ϵ, $\epsilon^2/(akT)$, and κa, where κ is the Debye reciprocal length, given by

$$\kappa^2 = 8\pi\epsilon^2 c/(DkT) \tag{31}$$

For aqueous latices, the new experimental variables q and c can be varied over wide ranges, with negligible variation in D (and none in ϵ). Thus the number of fully independent variables in the problem of the electroviscous effect is two more than in the corresponding hard-sphere problem.

Of the two principal studies reported on the second electroviscous effect, that of Brodnyan et al. suffers from incomplete characterization of particle size, size distribution and surface charge, while that of Wang does not extend to the highly interesting region of low electrolyte levels. An investigation combining the virtues of these two efforts is needed.

The optical diffraction studies of Hiltner and Krieger[41-43] have shown that the particles in a deionized monodisperse latex occupy the points of a close-packed lattice, in which the center-to-center particle distance D is given by

$$\phi(D/2a)^3 = 0.74 \tag{32}$$

Thus a deionized latex is an isotropic liquid crystal, stabilized by Coulombic forces. Such a crystal should show a yield point, which may be detectable only through measurements at ultra-low shear stresses. Also, a rheo-optical effect should be observable as the lattice is progressively destroyed by increasing shear rates, whereby the diffraction peaks observed by Hiltner and Krieger disappear completely at high rates of shear.

SUMMARY

Monodisperse latices come very close to realizing in the laboratory the models proposed by theoreticians to represent colloidal dispersions. When Coulombic and van der Waals forces are negligible, we have the hard-sphere dispersion of Einstein, Simha[44] and Mooney; with the inclusion of interparticle forces, the model spherical colloid is achieved.

New experimental methods permit us to prepare latices of high monodispersity, in a wide range of sizes, with controlled surface charge, and suspended in media of controlled electrical properties. Even prior to the perfection of these methods, latices were extensively used to test rheological theories. With the modern techniques, a new generation of definitive experiments is possible, exemplified by the establishment of a principle of corresponding rheological states for the steady flow of hard-sphere dispersions. In the near future, one can look forward to increased use of monodisperse latices in investigations of thixotropy, of electroviscous effects, and of the rheological consequences of heterodispersity.

ACKNOWLEDGEMENT

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